CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 11: Geometric Progression

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Find the common ratio and the 6th term of the G.P. $5, 10, 20, \ldots$
2. Find the 7th term of the G.P. $64, 32, 16, \ldots$
3. Which term of the G.P. $3, 12, 48, \ldots$ is 3072?
4. The 3rd term of a G.P. is 18 and the 6th term is 486. Find the first term and the common ratio.
5. Find the sum of the first 6 terms of the G.P. $2, 6, 18, \ldots$
6. Find the sum of the first 5 terms of the G.P. $16, 8, 4, \ldots$
7. How many terms of the G.P. $1, 3, 9, \ldots$ add up to 364?
8. Find the geometric mean of 9 and 16.
9. Find x if $x + 1$, 6 and $x + 6$ are in G.P. and x is positive.
10. The product of three numbers in G.P. is 1000 and their sum is 35. Find the numbers.

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Answer Key

1. $r = 2; t_6 = 160$
2. $64 \times \left(\frac{1}{2}\right)^6 = 1$
3. $4^{n-1} = 1024 = 4^5$; the 6th term
4. $r = 3, a = 2$
5. $\frac{2(729 - 1)}{2} = 728$
6. 31
7. $3^n = 729$; 6 terms
8. 12
9. $(x + 1)(x + 6) = 36$ gives $x = 3$
10. $a = 10; 2r^2 - 5r + 2 = 0$; 5, 10, 20

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